Short-term things to fix:

- Delete all of the writing that is unnecessary. ***Every time you talk about something, ask yourself "does an economist need to know this to understand the paper"?***
- Rewrite the introduction, and tell the ***clear story***:
  - Tech that increases the pace of productivity should be good right?
  - Except if it leads to a more competitive market, and firms want to undercut each other. What if they find a way to undercut each other that actually abuses a public resource?
  - In the livestock genetics market, the genetic strength of the pool is the public resource. Breeders can choose to abuse it for short-term gain by drawing too much from a certain pool, leading to disastrous consequences.
  - https://biology.ucdavis.edu/news/how-genetic-mutation-1-bull-caused-loss-half-million-calves-worldwide
  - The invention of genomic testing was an opportunity for breeders to find more productive genetic resources outside their gene pool. ***So why didn't they?? Why did the opposite happen?***
  - The basic theory: genomic testing led to a decrease in the generational interval, which drastically increased the speed of genetic progress. Under pressure to deliver higher gains, they chose to pull ***more*** from the most popular lines of bulls.
  - But what if they just did this because they were the most productive?
  - We test this hypothesis by constructing a counterfactual line of bulls that possessed the same quality of traits but were less popular at the introduction of genomics.
  - We find that inbreeding increased disproportionately more for the popular group. Companies did not invest in the most productive bulls, but instead the most popular, possibly in an attempt to gain market share.
  - This caused XXXX dollars of damage to the industry.
- Add a literature review, at least two paragraphs in the introduction
  - What work has already been done?
    - 90% of your citations should be ***ECONOMICS PAPERS***
  - How does this paper contribute to the literature?
- Ditch all of the theory models, write a conceptual framework **in words** that explains:
  - The incentives of the breeder to inbreed. Why would they do it? What's in it for them?
  - What genomic testing did. How did it impact how they make decisions?
  - Why did genomic testing increase the incentive to inbreed? Why would it not lead to breeders finding new and more diverse sets of genetics?
  - Why it is a "network externality"? What does it matter that it's a network externality?
- Empirical stuff
  - Balance test of your control and treatment group. Are they equivalent in terms of PTAs?
- Don't worry about policy recommendations. Talk about future work instead.

Long-term things to fix:

- I'm no longer sure that "network" externality is a useful concept here. This is basically a story about companies reacting to pressures pull from a public resource because of their short-term incentives.

- Similarly, it seems like "information" is not as important here. What really happened is that genomics made the pace of improvement basically double, and this means that the incentive was to do the thing that would bring them the most gains to beat everyone else. This happened to be inbreeding.
- A better theoretical framing for this is just from the perspective of dairy breeders (no demand side). One framing I've experimented with:
    - Dairy breeders have two choices: pull from the foreign pool or pull from the domestic pool.
    - The domestic pool gives them higher gains but contributes to inbreeding. The foreign pool gives them relatively lower gains but diversifies the pool.
    - ***The optimal social planner would choose to pull from the foreign genetics pool just enough to keep inbreeding under control.***
    - Each period, they pull from either pool. Inbreeding degrades the pool on aggregate, but each individual contribution to inbreeding is very small.
    - With a large enough N, the costs of inbreeding go to zero: each breeder's actual contribution to the erosion of the pool is so small that their individual actions don't matter to them enough to do anything about it.
    - As a strategic game, this would lead to everyone to choose the domestic pool as a best response. No one will willingly take on the burden of diversifying the pool, especially if this loses them market share.
    - If you did a simulation, you would probably find that everyone would inbreed until the population collapsed.
- All of the above requires a lot of work, but could be calibrated with a lot of functional forms from the dairy science and quantitative genetics literature.
- How do people in the seed industry deal with inbreeding? Do they have any insights on how to do this in the dairy genetics market?